

sparam_to_tdr.py

A Program for S-Parameters to Time Domain Reflectometry (TDR) Conversions

Motivation: Many Linear Circuit Simulators, EM Simulators and Vector Network Analyzers (VNA's) produce S-parameter files of the network being analyzed, but have no way to give a TDR result. This program can read standard Touchstone Format SnP files and calculate a TDR result for you.

If you aren't sure of how S-Parameter to TDR conversions work or their application see reference [1]. For more information on how to use scikit-rf, which does all of the heavy lifting calculation work here, see reference [2].

For Windows Install instructions see the section titled: “*Installation*” at the end of this document.

This document gives an overview of how to use the: “sparam_to_tdr.py” GUI.

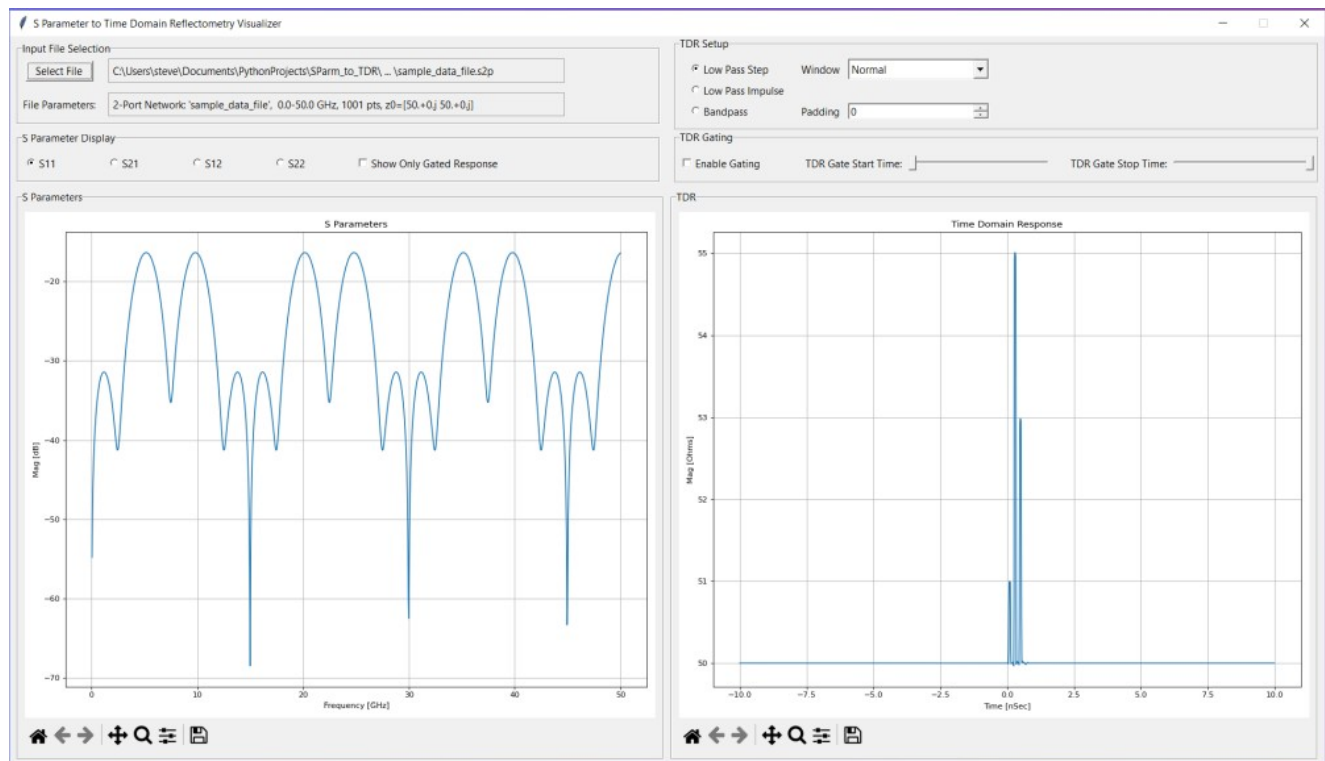


Figure 1 - The sparam_to_tdr.py Graphical User Interface (GUI) overview

1) Launch the program: `sparam_to_tdr.py`

2) A S2P Touchstone file must be loaded first. In the upper left hand corner of the GUI is the “Select File” button. Press this and a standard open file dialog will open. Select a valid S2P file. See figure 2. The sample data file provided in the Github repository will be used for the following examples (see: ‘sample_data_file.s2p’).

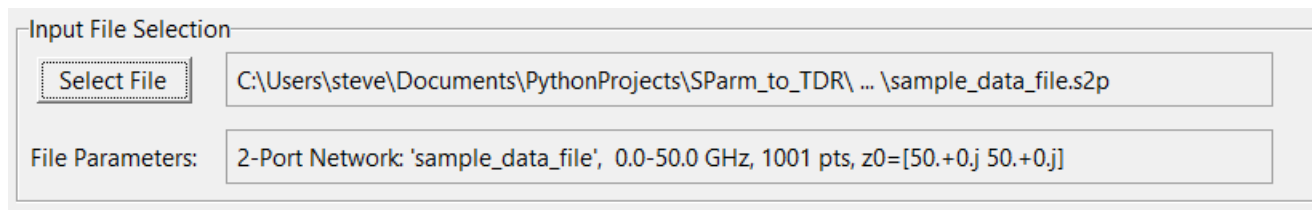


Figure 2 – In the upper left hand corner of the GUI is the input file selector button. Use this to select the desired Touchstone format S2P file to analyze. When a valid S2P file is loaded the file parameters are shown below the file file field.

3) When the file is selected it will be read in and processed by the scikit-rf Touchstone file reader. If successful then the network parameters will be displayed in the box to the right of the label marked: “File Parameters”. This field shows the file name, the frequency range, number of frequency points, and the reference impedance of the network (Z0). See figure 2.

3) immediately upon loading a valid network, the two graphs will show the selected S-parameter display on the left and the TDR calculation graph on the right. See figure 3. Each graph is a standard Matplotlib graph and includes a standard Matplotlib navigation toolbar. For information on how to use the navigation toolbar see,

<https://matplotlib.org/stable/users/explain/figure/interactive.html#interactive-navigation>

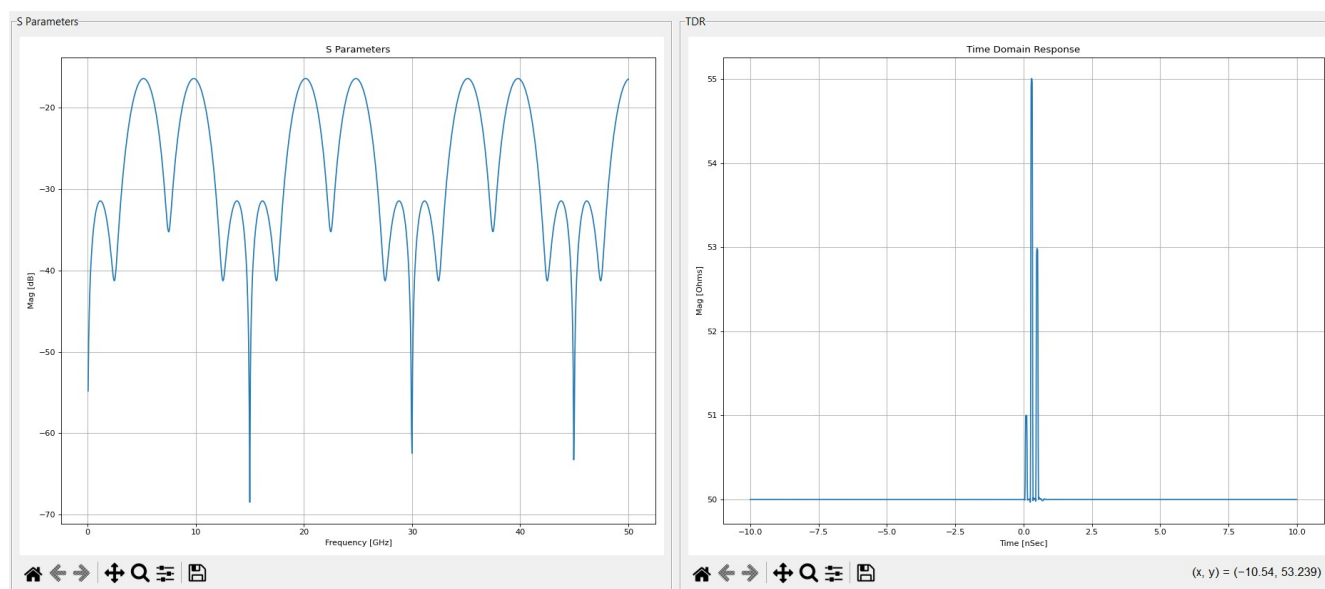


Figure 3 - Typical S-parameter and TDR graph displays. Below each graph are the standard Matplotlib Navigation Toolbars.

4) The S-Parameter to display is located just above the S-Parameter Graph. Valid selections are S11, S21, S12 and S22.

NOTE: Even though you can load in a Touchstone file with many more ports than 2, the display will only plot the first 2 port network that is in the Touchstone file.

5) On the TDR display usually the desired response will be a small portion for the calculated total TDR time-span. To 'zoom in', look below the TDR graph and select the 'Magnifying glass Icon' then draw a rectangle around the desired portion of the graph to get the interesting details. After 'zooming' the graph will remember the X axis scale factor and retain it until you zoom out. The Y axis will auto-scale with every change as appropriate. See figure 4.

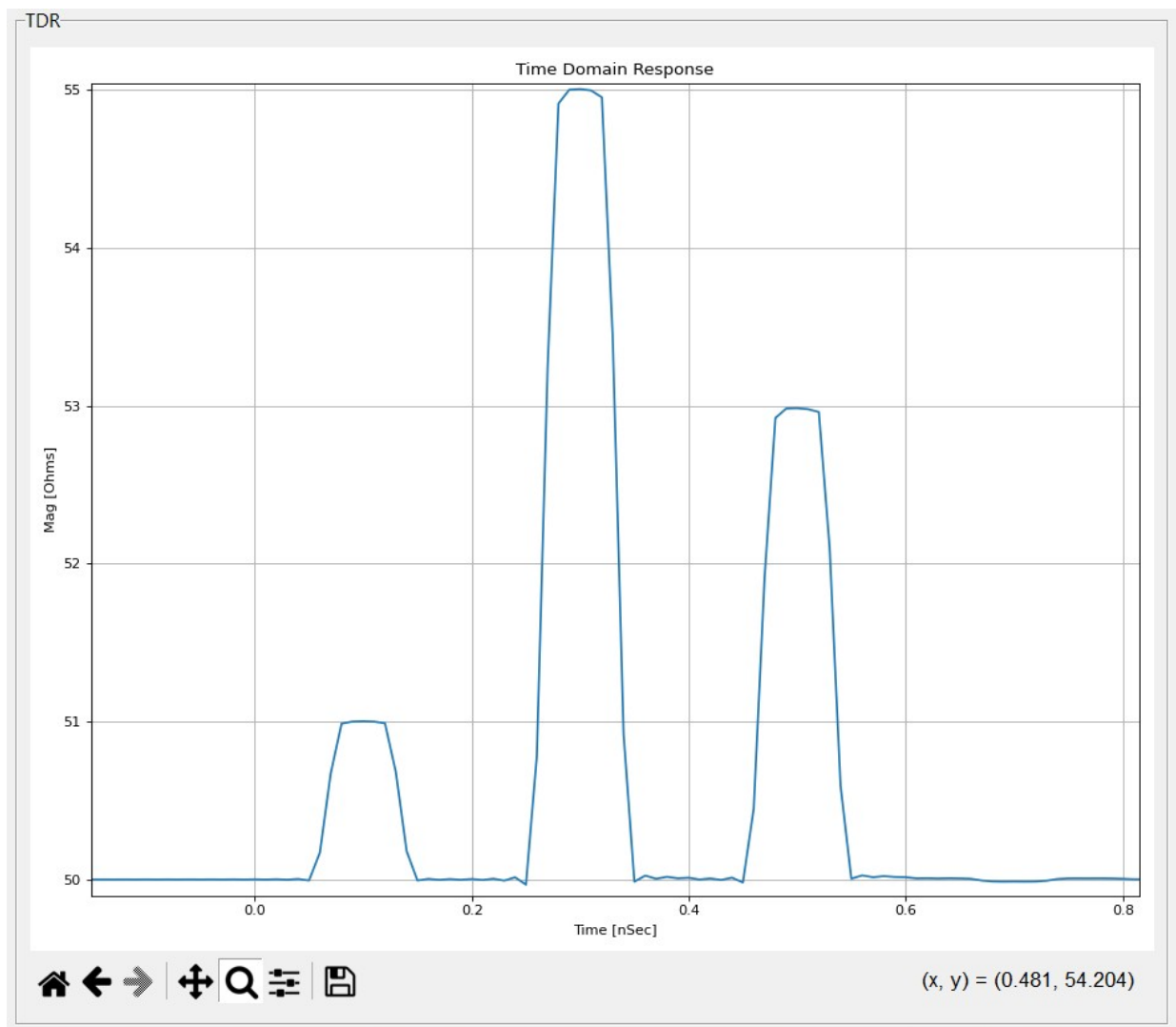


Figure 4 – Zooming in on the TDR plot is usually necessary to view the specific details of the TDR response.

6) Many of the TDR conversion parameters are adjustable. See the frame in the upper right corner of the window (figure 5). The TDR conversion type can be selected between Step, Impulse and Bandpass.

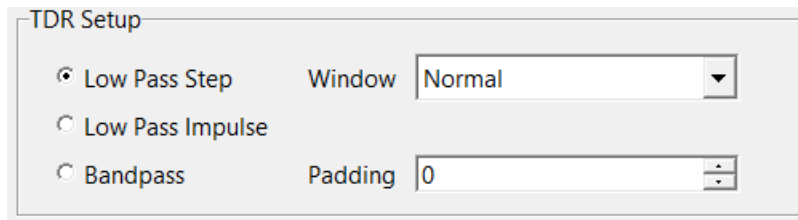


Figure 5 – TDR Conversion Parameters. The type of TDR response can be selected from: Step, Impulse and Bandpass. Additionally the Window and IDFT Padding values can be changed.

7) If very few data points are included in the S2P file, the Window can be changed to 'Maximum' and / or the Padding can be changed to improve the display.

The frequency range for Step and Impulse must extend to zero Hz for a proper display and this program automatically extrapolates the S-Parameter frequency to make this true. In Bandpass mode the frequency range does not have to extend to zero and whatever frequency that the S-Parameter file has is used unchanged.

'Window' can be 'Minimum' (which is really: No Window), 'Typical', or 'Maximum'. The same Kaiser windows as specified in reference 1 are used by this program.

'Padding' can be added to the Inverse Discrete Fourier Transform (IDFT) as required. This can help to 'fill in' the graph and make jagged peaks look more smooth. To adjust the padding simply scroll the mouse wheel over the field and the padding will change in 1000 step increments.

By adjusting the Window and Padding a suitably good looking display can be obtained. For the sample data file provided, adjusting the Window to: 'Maximum' and the 'Padding' to around 5000 makes the TDR graph much more pleasing to look at. See figure 6.

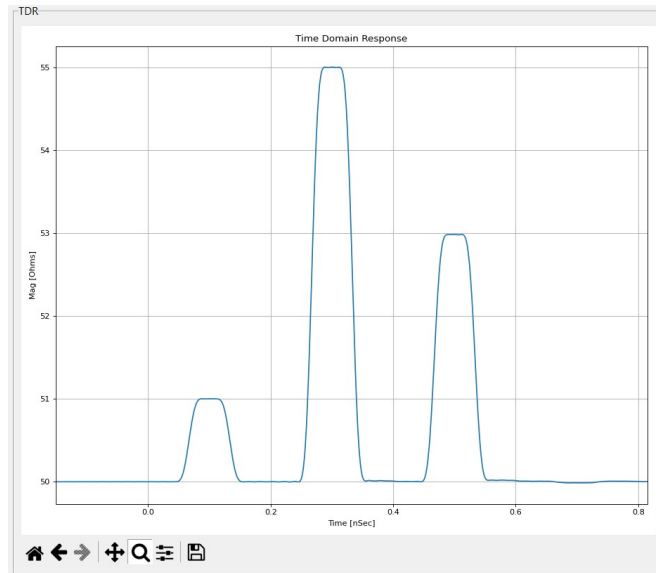


Figure 6 – By adjusting the ‘Window’ and ‘Padding’ the normal ‘jagged’ edges of the TDR peaks can be made more ‘rounded’ and ‘pleasing’.

8) Scikit-rf also provides the ability to ‘Gate’ or exclude portions of the TDR to give an approximation of the S-Parameter response as if that portion gated out does not exist, See reference [1] for more information on gating.

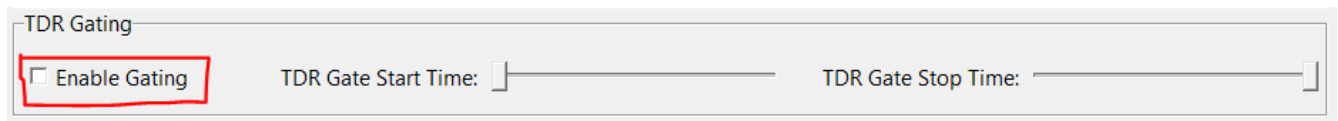


Figure 6 – To enable ‘TDR Gating’ click the checkbox as shown above. The sliders can be used to set the Gating start and stop times.

To use ‘Gating’ click the ‘Enable Gating’ Checkbox. See figure 6. The ‘Gate Start’ and ‘Gate Stop’ times can be adjusted with the sliders. The start is marked by a green vertical line, the stop time is a red vertical line and the center time is a dashed blue line. Generally when looking to make an analysis of a specific TDR feature, the center should be placed over that feature. See figure 7.

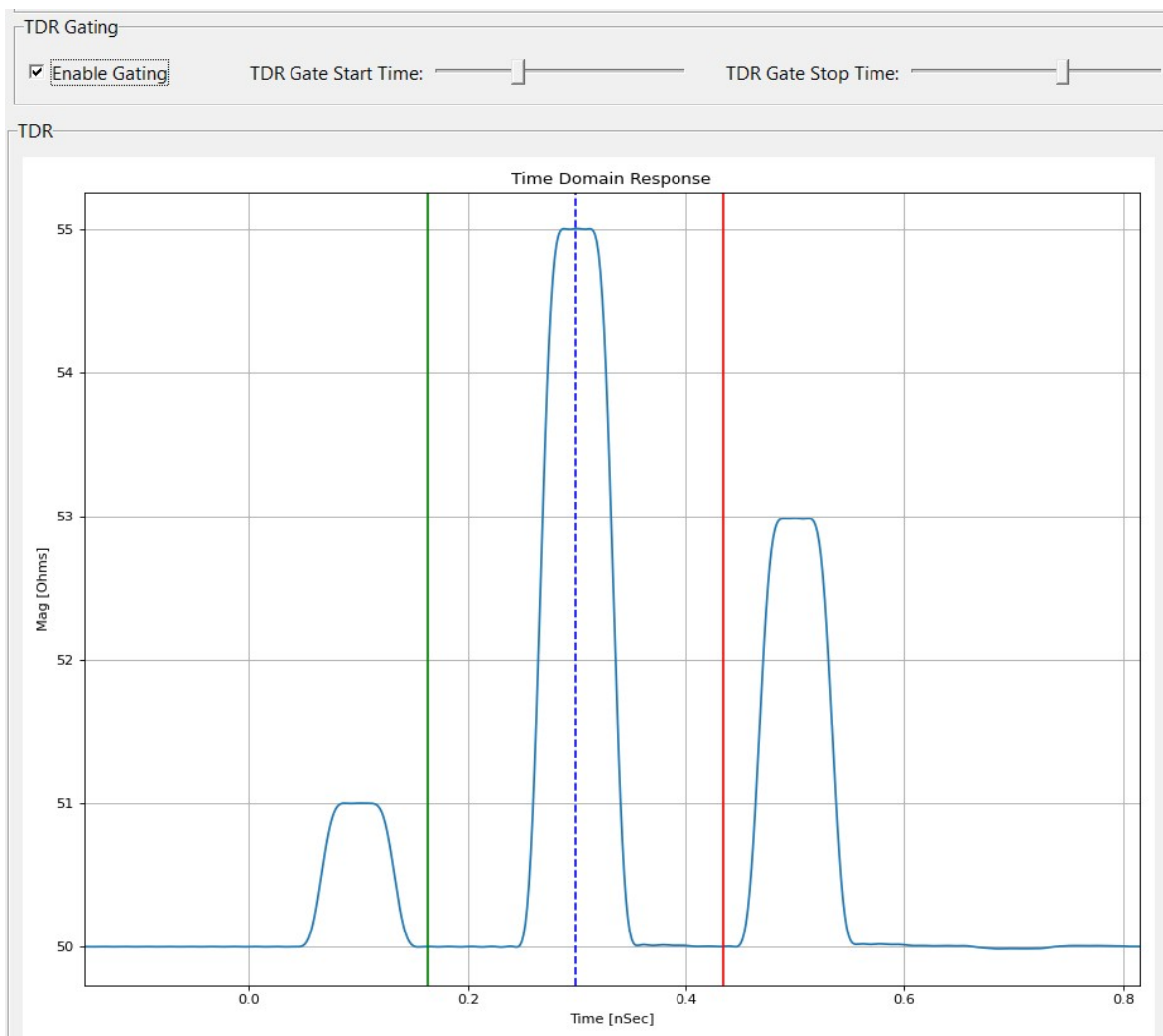


Figure 7 – Here the TDR Gating has been enabled and the Time Start and Time Stop Gates have been set to remove the response of the first and last ‘peaks’. Gate Start and Stop times are adjusted using the sliders above the TDR Graph.

9) The purpose of ‘Gating’ is to estimate what the S-Parameters would be if the start and stop impedances were removed. This is shown on the S-Parameter display by adding a second trace in Red that shows the estimated or ‘gated’ response along with the original response as shown in figure 8.

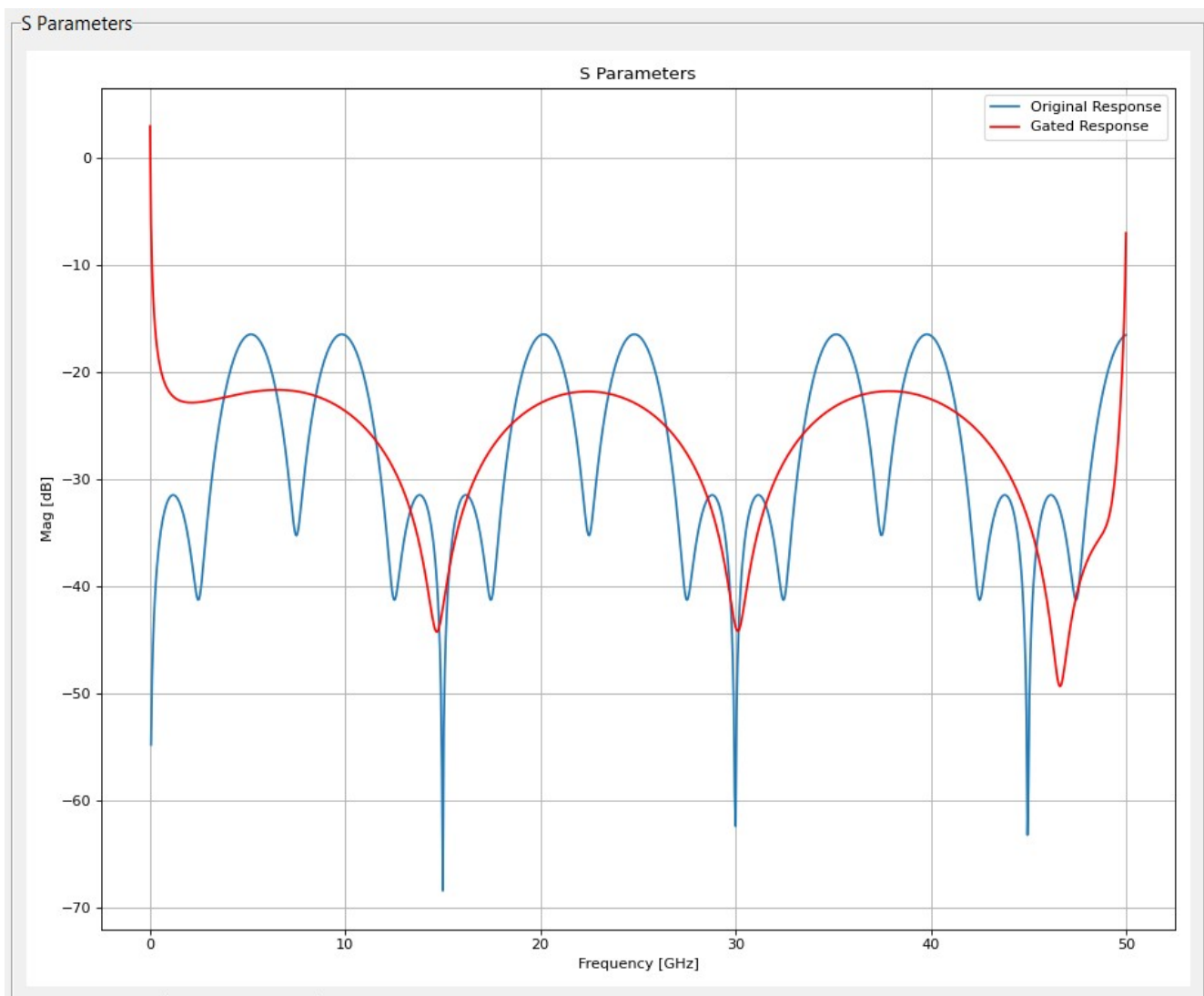


Figure 8 – When ‘Gating’ is applied the S-Parameter graph changes to show the original S Parameter and the estimated Gated response. If only the gated response is desired, then the “Show Only Gated Response” checkbox above the graph can be checked.

Installation for Windows:

System Requirements:

Tested on Windows 7, 10 and 11 only.
Screen resolution of at least 1920 x 1080 (FHD).

Install Procedure:

1) From the Github Repository, get the files in the 'src' directory. Place the files in some folder on your PC.

File descriptions:

sparam_to_tdr.py - The main application file - run this file!

sparam_to_tdrui.py - The GUI Loader File. This file is autogenerated by PyGuBu. You need this file to run the application.

sparam_to_tdr.ui - The XML representation of the PyGuBu GUI design. You need this file to run the application. This file is autogenerated by PyGuBu. You can also use this file to modify the GUI in PyGuBu.

sample_data_file.s2p - This is a sample s-parameter file that can be used to test the functionality. This file is optional.

requirements.txt - The standard Python dependencies list. Use this to make sure that the proper libraries are added. Only needed to get proper libraries, not needed at runtime.

2) Make sure that Python 3.12 or greater is installed on the computer.

3) Use the requirements.txt file to get the proper program dependencies.

Open a command prompt at the location where the requirements.txt file is, and run the command: `pip install -r requirements.txt`

4) Double click on the file: `sparam_to_tdr.py` to start the program.

References:

[1] "Time Domain Analysis Using a Network Analyzer", Keysight Technologies. Document number: 5989-5723

[2] Scikit-rf Time Domain and Gating:

["https://scikit-rf.readthedocs.io/en/latest/examples/networktheory/Time%20Domain.html"](https://scikit-rf.readthedocs.io/en/latest/examples/networktheory/Time%20Domain.html)

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